their position, as Michels pointed out). To avert this threat and fortify their position, union leaders instituted various services to members that would make them and the local again important to the membership.[21] A final example of the succession of goals is furnished by a voluntary association. On the basis of his study of a large service organization staffed by volunteers—the National Foundation for Infantile Paralysis—Sills correctly predicted that this organization would turn to new programs with the accomplishment of its original objectives.[22] This prediction was based on the facts that the organization had received wide public acceptance, had successfully attained its initial objectives, and was governed by a corporate-type structure that had the capacity and interest to establish new policies as well as the power to implement them.

What conditions govern whether the displacement of goals or the succession of goals is the prevailing trend in an organization? A crucial factor seems to be the organization's relation to its environment. As long as its very survival is threatened by a hostile environment, its officers will seek to strengthen the organization by building up its administrative machinery and searching for external sources of support. This process is often accompanied by a retreat from the original goal to more modest objectives, as exemplified by the history of the Tennessee Valley Authority and by the tendencies in socialist parties and unions in imperial Germany. But if the community permits an organization to succeed in achieving its initial objectives, the staff's interest in preserving the organization and expanding its jurisdiction will lead to the advocacy and adoption of more advanced goals, as illustrated by the federal agency, the Windsor local as well as other American unions, and the National Foundation. This explanation, however, does not account for the rigid conformity with procedures found in many old bureaucracies that are not at all threatened by the community. Perhaps the environment must supply stimulating challenges as well as support to organizations for flexibility and the succession of goals to develop, just as both these conditions seem to be needed for individual creativity (see the study by Pelz discussed in Chapter V).

Is it correct to speak of increased bureaucratization in those organizations where original goals are succeeded by more advanced ones? Yes, in the sense that the change involves increased scope and power for the bureaucracy; no, in the sense that it involves less preoccupation with administrative procedures as ends-in-themselves. The strain is toward innovation, not rigidity, and in this sense represents

21 Lipset *et al. (op. cit.,* p. 407) have noted that an important latent consequence of the proliferation of activities in unions is to encourage more active participation by members.

22 David L. Sills, *The Volunteers,* Glencoe, Ill.: Free Press, 1957, pp. 253–268.

a debureaucratizing tendency. We turn now to a discussion of some other forms of debureaucratization.

Debureaucratization. Less attention has been devoted to processes leading to reduced bureaucratization than to those leading to increased bureaucratization. A recent article by Katz and Eisenstadt analyzes some cases of debureaucratization reported in previous studies, most of which deal with tendencies to relax hierarchical authority.[23] It has been observed, for example, that army discipline, strict lines of authority, and social distance between officers and men were less pronounced in combat than in peacetime, particularly if the combat unit was isolated from other units. A study of an industrial concern found that miners were able successfully to resist attempts by management to increase bureaucratization while factory workers in the same company could not. Research also indicates that men on the night shift in industrial plants were less subject to hierarchical authority and discipline than men on the day shift. Katz and Eisenstadt suggest that the common elements in these situations are the presence of physical danger and the isolation of the unit from the larger organization. Both of these conditions make superiors in some respects dependent on their subordinates, and their dependence constrains them to refrain from using authoritarian or coercive measures in performing their duties and to rely, instead, on more personal, nonbureaucratic means of motivating cooperative effort. Katz and Eisenstadt infer from their secondary analysis that dependence of officials on their subordinates promotes debureaucratization, and that this principle applies to dependence on clients as well as on subordinates in an organization. They further extend this principle by following a lead of Parsons and including in the concept of dependence not only the fact that a client or subordinate has the power to influence the life-chances of an official but also the mere fact that the former is capable of disrupting the role performance of the latter.[24]

Data to illustrate this conclusion are gathered by Katz and Eisenstadt in their study of the contacts between immigrants and bureaucrats in Israel. Official-client relations depend on the abilities of both parties properly to perform their roles. Since immigrants who came to Israel from non-Western countries typically did not know what was expected of a bureaucratic client, their inability to act appropriately in this role disturbed the official-client relationship. For

[23] Elihu Katz and S. N. Eisenstadt, "Some Sociological Observations on the Response of Israeli Organizations to New Immigrants," *Administrative Science Quarterly*, 5 (1960), pp. 113–133.

[24] Parsons notes that although the parents have power over the life-chances of the child, they are also dependent on the child, because he or she is capable of disrupting the family system. See Talcott Parsons and Robert F. Bales, *Family, Socialization and Interaction Process*, Glencoe, Ill.: Free Press, 1955, pp. 46–47, footnote 18.

example, immigrants sometimes tried to bargain with the bus driver over the amount of the fare or to debate the route the bus was to travel. In such situations, an official was obliged to teach his clients how to play their role before he could properly perform his. Again, the public-health nurse had to teach her prospective clients which of their problems she could deal with and which ones were outside her sphere of competence. But the diffuse relations with clients that developed as officials assumed the role of teacher and counselor tended to impinge on their specific roles as representatives of a given bureaucracy. It led to debureaucratization and occasionally to entirely new role relationships independent of the initial bureaucratic ones. Some of the instructors sent by the government to immigrant villages, for instance, came to assume boundary roles mediating between the government and "their" village and, in the extreme case, turned into representatives of the immigrant group or even into leaders of movements furthering this group's interests.

The hypothesis that the dependence of officials on clients (or subordinates) lessens the degree of bureaucratization can be tested with data from County Agency previously presented in another connection. It will be recalled that in our comparison of the Public Assistance Division and the Child Welfare Division of County Agency it was noted that the clients of the latter were not only children but also foster parents, on whose cooperation operations in CWD were dependent, while there was much less dependence on the recipients of public assistance in PAD. The prediction is, therefore, that CWD should be less bureaucratized than PAD. The findings support this conclusion. Thus, caseloads were smaller in CWD, and there was more opportunity for casework in this division. Assignment of cases was less bureaucratic in CWD—not by geographic area, as in PAD, but by matching the severity of the problem with the experience of the worker. Since no case reassignments were necessary in CWD when the client changed residence, worker-client relations were more stable and workers were not considered to be interchangeable. Finally, the feelings of the client about the race of the worker were taken into account in CWD but not in PAD. (Note that lack of bureaucratization, even if in the interest of giving professional considerations wider scope, opens the door to racial discrimination.) These differences conform to the prediction. With only two cases, however, one cannot say with confidence whether these differences in bureaucratization actually were the result of the greater dependence of CWD on clients or were due to other conditions, such as the fact that CWD served children or that it had legal responsibility for their care.

Diamond presents an extreme case of debureaucratization in his

discussion of an organization that became a society.[25] The Virginia Company was established in 1607 as a business concern to exploit the riches of a new continent and the labors of native peoples. Virginia, however, provided conditions different from those that had been faced by similar companies in India, Mexico, and Peru. Native labor refused to be mobilized, and mineral wealth was not to be found. Hence, it proved necessary to establish an agricultural community based on imported labor, and the company had to devote its efforts to recruiting the necessary voluntary labor from England. After these laborers had arrived, they were subjected to strict discipline enforced by a military regime and by religious sanctions. However, not enough men responded to the call, and the Company was forced to offer more and more concessions to serve as inducements for migrating. The terms under which the laborers could obtain land were eased, women were sent to Virginia to become the wives of the settlers, and martial law was gradually limited as representatives of the settlers were given some voice in government. As a consequence of these changes settlers became involved in a complex network of statuses, many of them outside the organization of the Virginia Company.

At one time in Virginia, the single relationship that existed between persons rested upon the positions they occupied in the Company's table of organization. As a result of the efforts made by the Company to get persons to accept that relationship, however, each person in Virginia had become the occupant of several statuses, for now there were rich and poor in Virginia, landowners and renters, masters and servants, old residents and newcomers, married and single, men and women; and the simultaneous possession of these statuses involved the holder in a network of relationships, some congruent and some incompatible, with his organizational relationship.[26]

Gradually, settlers felt that their statuses outside the Company were the more important, and they were no longer willing to accept organization position as the primary basis of legitimate authority. Thus, a society emerged where before there had been only a formal organization, a transformation that constitutes the polar case of debureaucratization.

Emergent Patterns

Informal Organization. The formal organization sets the stage for the emergence of informal social processes. Much emphasis has been placed on the influence exerted by the emergent informal relations and group

[25] Sigmund Diamond, "From Organization to Society," *The American Journal of Sociology*, 63 (1958), pp. 457–475.
[26] *Ibid.*, p. 471.

norms as they mold the patterns of conduct of the members of the organization. But formal institutions help to shape the characteristics of the informal organization, and this process has received far less attention than it merits. This latter process is exemplified by our analysis of the implications of the presence of a procedure manual for consultation networks in the two welfare agencies.

Oldtimers were more likely to be respected for their competence as caseworkers than newcomers in both agencies, and respected workers were more likely than others to be regularly consulted by colleagues in both. But in City Agency, where there was no procedure manual, seniority exerted an additional independent influence on a worker's participation in consultation, whereas it did not in County Agency, where all workers possessed a procedure manual. Only in City Agency did we find that oldtimers were disproportionately often consulted, and that newcomers consulted others disproportionately often. These relationships were not affected by holding constant the respect a worker's competence commanded. Worrying about the work, on the other hand, was associated with consulting colleagues frequently in County Agency but not in City Agency. The absence of a procedure manual apparently puts a premium on experience in the organization, with the result that seniority as such, regardless of whether it is accompanied by superior competence or not, has much impact on the patterns of informal communication among workers. In the presence of a procedure manual, which serves as a substitute for personal experience and reduces differences in the objective need for advice between workers, the subjectively felt need for approval and support, as indicated by worrying, exerts more influence on the tendency to seek advice. Here we see how an aspect of the formal institutions—the existence of a procedure manual—affects the informal relations that emerge among peers and, specifically, the significance that various characteristics of workers have for their informal status in the work group.

Another example of the influence of conditions in the formal organization on informal processes is provided by the finding from the study of City Agency that work pressures promoted the development of consulting relations that were perceived as reciprocal. Such reciprocity in a group's consultations conditioned the relationship between the pressures on individual workers and their consultation practices. Thus, if high reciprocity in the work group made consultants easily accessible, the greater the work pressure the more often did the individual consult with others. But in low-reciprocity groups, where consulting involved more effort, the greater the work pressure the less often did the worker consult his fellows.

If the formal structure influences the development of informal patterns, this emergent group structure, in turn, influences operations.

One important aspect of the informal structure is group cohesiveness. Earlier studies have shown that cohesion determines the degree of control that the group can exert over its members. Our research indicates that high cohesion made workers more independent of and more detached toward clients. Group values and norms, too, influence operations. We found that the work group's prevailing orientation to clients influenced a worker's approach to clients irrespective of what his own orientation was. Value orientations also affect the group norms that develop. For example, proclient values in groups seemed to give rise to the norm that clients should be given more independence, which helped protect clients against any individual worker who might not share these values. Group norms often regulate productivity. Whether they encourage a high or low level of productivity apparently depends on whether maximum productivity corresponds to or conflicts with the values and interests of the work group. If economic conditions are bad and workers fear layoffs, and if competition threatens group solidarity, then group pressures tend to restrict output. In the absence of such fears, however, and if a cooperative task makes workers dependent on one another and averts the danger of competition, group pressures tend to increase output. The former situation prevailed in Roethlisberger and Dickson's Bank Wiring Observation Room, the latter, in Walker's automated steel mill.[27]

Conformity to group pressure is associated with informal status. When the central values and norms of the group are involved, only outcasts are apt to deviate, and deviant tendencies are penalized by loss of informal status in the group. But in less salient areas, such as the prevailing climate of opinion in a group, the better integrated person seems to be more capable of resisting group pressures than the one who has still to win full acceptance. Informal status was also found to influence the orientations of workers to both their clients and their colleagues. Newcomers were more likely than oldtimers to be procedure-oriented rather than service-oriented in their approach to welfare clients, presumably because a lack of familiarity with procedures engenders anxiety and rigidity. Social support from peers seems to have mitigated such anxiety, as was indicated by the finding that newcomers who were socially integrated were just as apt to be service-oriented as oldtimers and much more so than unintegrated newcomers. With regard to colleagues, newcomers who were not yet accepted among their fellow workers were more concerned with being highly thought of by their peers than newcomers who had already achieved an integrated position among them. In contrast, oldtimers who were not highly inte-

[27] See F. J. Roethlisberger and William J. Dickson, *Management and the Worker,* Cambridge, Mass.: Harvard University Press, 1939, pp. 379–524; and Charles R. Walker, *Toward the Automatic Factory,* New Haven: Yale University Press, 1957, pp. 80–83.

grated among colleagues were less interested in their colleagues' opinions of them than oldtimers who had gained full acceptance. Apparently, the importance of peers as a reference group tended to increase over time for those workers who had achieved an integrated position but to decrease over time for those who had failed to attain a secure informal status. These are some illustrations of the internal dynamics characteristic of the emergence in work groups of informal organizations, which exert a pronounced effect on the operations in the formal organization.

Supervision. Emergent processes characterize authority relations as well as peer relations. Formal conditions and informal processes are even more closely intertwined in hierarchical relations than in peer-group patterns. Effective supervision requires extending the scope of authority beyond the narrow limits rooted in the formal contract. The successful supervisor must do more than maintain discipline and compliance with official orders—he must encourage his subordinates to exert effort, to assume responsibility, and to exercise initiative. His ability to obligate subordinates is of crucial importance for extending his influence over them and motivating them to follow his guidance and suggestions readily. The formal organization aids him in creating social obligations among subordinates. Thus, the promotion procedure is designed to assure that the supervisor has the competence to furnish subordinates with assistance in the form of advice and training. His position of greater power and centrality in the communication network enables him to extend a variety of services to subordinates—to shield them against demands of management, to obtain staff services for them, to back them in controversies with other departments, and many others. Formal rules and sanctions and his status prerogatives provide opportunities for obligating subordinates simply by refraining from enforcing or exercising them. While official institutions supply the superior with the tools for rendering services to subordinates, he may or may not be able to take advantage of these and actually succeed in obligating subordinates to him.

To establish authority over his subordinates, the supervisor must be able and willing to furnish services that command their respect and allegiance. For the collective loyalty of subordinates is what legitimates his exercise of control over them and transforms it into authority. When respect for the supervisor and feelings of obligation to him prevail in a group, they give rise to a consensus that, since it is in the common interest to maintain his good will, his directives and requests must be followed. Once these group norms enforce compliance with the supervisor's directives, his influence becomes independent of the use of coercive sanctions, or of persuasion, or even of the need to oblige particular subordinates in exchange for every request made of them. In

short, group loyalty is the core of the value orientation that legitimates the supervisor's exercise of authority over subordinates, inasmuch as it leads to the development of group agreement that his requests are to be followed voluntarily and of group constraints that enforce this social agreement.

The formal organization cannot create feelings of loyalty to superiors among subordinates, but it can create conditions that promote them. Hence while the superior's informal practices determine the actual scope of his authority over subordinates, formal institutions can encourage or interfere with the development of effective authority relationships. We have already mentioned several examples of how official procedures increase the chances that supervisors can and will render services to subordinates. Let us add one illustration of how the formal structure can discourage superiors from acting in ways that would undermine the loyalty of their subordinates and their authority. A flat structure—an obtuse hierarchical pyramid—by increasing the number of their subordinates, prevents superiors from supervising too closely and thereby alienating their subordinates. At the same time, this flat structure prevents them from leaning too heavily on their own superiors and thus losing their independence and the respect of their subordinates. Further, it makes it less likely that superiors become overinvolved with their subordinates, simply because there are so many of them, and it consequently promotes detachment.

The supervisor has often been described as "the man in the middle," a description which suggests the significance of his relations to levels both above and below him. Data from a variety of organizations, including the two welfare agencies, indicate that some social distance from both subordinates and superiors is important for effective supervision. For example, detachment from subordinates was found to be associated with high productivity, and independence from superiors, with greater solidarity in the work group. And both kinds of social distance, although the two were hardly related to one another, were associated with commanding the loyalty of subordinates. In seeking to ascertain the source of the social support that permits a supervisor to remain independent from his superior and detached from his subordinates, we hypothesized that the supervisor's own peer group serves this function. Empirical findings negated this hypothesis, however, probably because the social relations among supervisors on the same level were not salient for them in the organizations studied. A possible alternative explanation is that the loyalty of subordinates might have provided the support needed for supervisors to maintain their independence and detachment.

Commanding the loyalty of subordinates and expressing loyalty to superiors were inversely related for City Agency supervisors. This dif-

ference suggests that superiors and subordinates function as alternative sources of social support for the supervisor. If he receives support from one source, he has less reason to seek it from the other. This finding (yet to be confirmed in research) raises an interesting paradox: if a manager commands the loyalty of his subordinates, who are supervisors, they will be under less pressure to seek to win the loyalty of their own subordinates, a condition which would be dysfunctional for their effectiveness. Conversely, a manager's inability to command the loyalty of his subordinates would have important latent functions for their effectiveness as supervisors, since it would constrain them to try to earn the respect and loyalty of their subordinates and thus foster independence from the manager and detachment toward subordinates. These speculations would imply that effective supervision can be expected only on alternate levels in the hierarchy, not on all levels, and (if they prove to be correct) they may explain in part why multilevel hierarchies have serious shortcomings.

Somewhat parallel conclusions can be drawn from two case studies of managerial succession. Gouldner reports on a new top manager who had recently been brought to a gypsum plant from outside and was under heavy pressure from the management of the company that owned the plant to improve production.[28] The new manager, in obvious contrast to his predecessor, was a stranger, unfamiliar with the network of informal relations in the plant, and he had as yet to earn the loyalty of the personnel of the organization. These typical characteristics of the role of successor constrained the manager to resort to bureaucratic mechanisms, such as elaborate rules and close supervision, in order to improve productivity. Such innovations alienated subordinates further. Since loyalty to him did not develop, the manager found it necessary to maintain control by further bureaucratization. Perhaps he felt too loyal to his superiors to permit him to win the loyalty of his subordinates. A recent case study by Guest deals with a very similar situation.[29] Again, a new manager was brought to a plant from the outside and put under pressure by the company who owned the plant to improve production. In this case, however, the new manager sought to achieve the same goal by opposite means. He tried, successfully, to raise productivity by relaxing the bureaucratic rules that had been set up under his predecessor and by refraining from using formal sanctions, thereby promoting loyalty.

Why did the two managerial successors use opposite strategies? A crucial difference between the two situations in which the new managers found themselves was that the predecessor in the gypsum plant

[28] Alvin W. Gouldner, *Patterns of Industrial Bureaucracy*, Glencoe, Ill.: Free Press, 1954, pp. 45–101.

[29] Robert H. Guest, "Managerial Succession," paper read at the meetings of the American Sociological Association, New York, 1960.

had commanded the high loyalty of his staff, whereas the predecessor in the second case had been generally disliked. The predecessor's practices appear to have served as reference standards for judging the new manager, which made it difficult for Gouldner's manager but easy for Guest's to win the approval of employees. Since the successor to an unpopular manager could rather easily earn the loyalty of subordinates, there was little need for him to resort to bureaucratic mechanisms to establish control over operations. From these and the foregoing conclusions, the following hypothesis may be inferred: a manager's ability to command the loyalty of his subordinates is inversely related to both his predecessor's and his superior's ability to command loyalty.

Conflict and Change. There seems to be a relation of mutual dependence between conflict and change in formal organizations: changes in the social structure often precipitate conflict, and conflicts tend to generate innovations.[30] Thus, conflicts between bureaucrats and clients frequently give rise to new practices which help to resolve or mitigate such disturbances, as exemplified by the custom of complaining and joking about clients to colleagues. This practice tended to reduce tensions and improve operations, albeit at the cost of less considerate treatment of clients. In those situations where clients are a part of the organization rather than simply isolated persons in occasional contact with it, conflicts between them and officials affect their social structure as well as that of the officials. An illustration is the elaborate network of relations that develops among prisoners as they attempt to gain some control over their environment in opposition to the formal system. This emergent organization among prisoners affects and is affected by the organization of the officials, particularly the guards. The connective tissues linking staff and client groups in mental hospitals appear to be extremely sensitive, inasmuch as even covert conflicts and disagreements among the staff have important repercussions among patients.

Bureaucratic formalization is one method for reducing uncertainty in formal organizations. Official procedures provide precise "performance programs,"[31] which prescribe the appropriate reactions to recurrent situations and furnish established guides for decision-making. But uncertainty cannot be completely eliminated. Change, whether due to new external developments impinging on the organization or to internal modifications, produces situations without established precedents. Besides, some exigencies that may arise cannot be anticipated,

[30] These statements apply to two distinct types of conflict: (1) conflict between principles that govern organizational conduct—a good example is the conflict between providing services to clients versus checking their eligibility for assistance; and (2) conflict between persons or groups, which, of course, is sometimes due to conflicts between the principles that govern their performance.

[31] On this concept, see James G. March and Herbert A. Simon, *Organizations,* New York: Wiley, 1958, pp. 141–142.

and the more complex the tasks, the more do attempts to prescribe behavior too rigidly interfere with the ability to make expert judgments in terms of abstract professional standards. Finally, procedures that lessen uncertainty for one group may increase it for another. For example, Gouldner points out that short-term employment contracts increase management's ability to deal with fluctuations in the market but at the same time reduce the certainty of employees concerning their job situations.[32] Informal patterns typically develop precisely in these areas of uncertainty and are attempts to cope with them. Where the uncertainty is one-sided, as when organizational policies increase the certainty of management at the expense of that of the workers, the informal organization of workers is likely to conflict with managerial objectives. If management policies deny employment security to workers, for instance, the workers' informal practices may take the form of restricting output to protect themselves against layoff and to reduce competition for scarce positions. But where the uncertainty is two-sided, not a handicap to one group and an advantage to another, informal patterns often arise that serve to organize operations in the interests of organization objectives.

Both Jaques and Dalton have noted that an individual's ability to stand uncertainty and ambiguity governs the scope of the responsibility he will seek.[33] Dalton suggests that strong managers, as opposed to weak ones, move into areas of uncertainty in the hope of extending their responsibility and influence. Power struggles develop as various managers seek to organize these pockets of uncertainty in the interest of their departments, and coalitions are formed to effect this organization. These power struggles, and the changes and uncertainties they bring, offer welcome opportunities to middle-level managers to extend their responsibility. The interests of strong managers in expanding their influence in such contests help the organization to adapt to change, since they cause these managers to welcome rather than to resist it. In the highly bureaucratized French tobacco monopoly, Crozier

32 Alvin W. Gouldner, "Discussion" of Wilbert E. Moore's "Industrial Sociology," *American Sociological Review,* 13 (1948), pp. 396–400. Abegglen presents a description of the opposite situation in Japanese factories, where a permanent employment contract increases the certainty of workers but makes management more insecure by handicapping it in its attempts to adjust to changing market conditions or to introduce new technologies. James C. Abegglen, *The Japanese Factory,* Glencoe, Ill.: Free Press, 1958, pp. 11–25.

33 Elliott Jaques, *The Measurement of Responsibility,* London: Tavistock Publications, 1956, pp. 85–106; and Melville Dalton, *Men Who Manage,* New York: Wiley, 1959, pp. 243–248, 252–255. Both Jaques and Dalton tend to view this characteristic—the individual's capacity to stand ambiguity—in psychological or sociopsychological terms whereas we would prefer to concentrate attention on the individual's position in the social structure as it influences his ability to cope with prolonged uncertainty.

found, power struggles developed in the one major remaining area of uncertainty—machine breakdown.[34] Conflicts occurred between maintenance and production personnel as the maintenance staff and, particularly, the managers in charge of it expanded their area of control and invaded that of line officials. Informal power accrued to the experts whose technical skills enabled them to cope with uncertainty. For knowledge of technical principles facilitates handling uncertainty in two ways: by making it possible to deal with emergencies, such as machine breakdowns, and by increasing the ability to stand uncertainty for prolonged periods.

Processes similar to those that govern conflicts within organizations characterize the conflicts between organizations. Thus, the coalitions and mergers of organizations to achieve dominance in a market can be looked on as a method of reducing uncertainty. Competition between firms increases the certainty of the public in the fairness of the prices charged but at the same time decreases the certainty of firms, which often react by trying to improve the security of their position by developing formal and informal ties with competitors. As uncertainties are reduced in one area (for example, between sellers in a common market) they arise in others (for example, between buyers and sellers), and the power struggle begins again in a new arena.

Dilemmas of Formal Organization

We shall review three dilemmas of formal organization: (1) coordination and communication; (2) bureaucratic discipline and professional expertness; (3) managerial planning and initiative.

Coordination and Communication. The experiments and field studies on communication and performance we have reviewed lead to the conclusion that the free flow of communication contributes to problem-solving. There are three ways in which decisions are improved by the unrestricted exchange of ideas, criticisms, and advice. First, social support relieves the anxieties engendered by decision-making. In the discussion of problems with others, their social approval of the first step taken toward a solution mitigates the anxieties that might otherwise create a blocking of associations, and it thus facilitates reaching a solution. Once consultation patterns have become established, moreover, the very knowledge that advice is readily accessible makes it less disturbing to encounter a difficult problem, and the experience of being consulted by others strengthens self-confidence; both factors lessen anxieties that impede decision-making.

[34] Michel Crozier, *The French Bureaucratic System,* Stanford: Stanford University Press (forthcoming).

Second, communication processes provide an error-correction mechanism. Different persons are guided by different frameworks in their approach to a given problem, and the differences make it easier for them to detect the mistakes and blind spots in the suggestions of one another. Although social support and error correction are in some respects opposite processes, both of them are, nevertheless, important for problem-solving, as indicated by Pelz's finding that optimum research performance is associated with consulting some colleagues whose orientation differs from one's own (who challenge one's ideas) and some who share one's orientation (who support one's ideas).[35]

Third, the competition for respect that occurs in the course of discussing problems furnishes incentives for making good suggestions and for criticizing the apparently poor suggestions of others.

While the free flow of communication improves problem-solving, it impedes coordination. Unrestricted communication creates a battleground of ideas; the battle helps in selecting the only correct or best among several alternative suggestions, but makes it difficult to come to an agreement; and coordination always requires agreeing on *one* master-plan, even though different plans might do equally well. Processes of social communication, consequently, make the performance of groups superior to that of indviduals when the task is finding the best solution to a problem but inferior when the task is one of coordination.

Hierarchical differentiation is dysfunctional for decision-making because it interferes with the free flow of communication. Studies of experimental and work groups have shown that status differences restrict the participation of low-status members, channel a disproportionate amount of communication to high-status members, discourage criticism of the suggestions of the highs, encourage rejecting correct suggestions of the lows, and reduce the work satisfaction of the lows and their motivation to make contributions. All these factors are detrimental to effective problem-solving. If hierarchical differentiation does not block but frees the flow of communication, however, it improves decision-making; this observation indicates that the adverse effects that hierarchical differentiation typically has for problem-solving are specifically due to the obstacles to free communication it usually creates. But the very restriction of communication that makes hierarchical differentiation dysfunctional for problem-solving improves performance when the task is essentially one of coordination. Experiments with various communication networks show that differentiation, centralized direction, and restricted communication are necessary for efficient coordination. However, the achievement of such a differentiated organization—

35 Donald C. Pelz, "Some Social Factors Related to Performance in a Research Organization," *Administrative Science Quarterly,* 1 (1956), pp. 310–325.

itself a problem-solving task—seems to have been easier for groups in which communication flowed freely than for those where it was experimentally restricted.

These conclusions point to a fundamental dilemma in formal organizations. Organizations require, of course, both effective coordination and effective problem-solving to discharge their functions. But the very mechanism through which hierarchical differentiation improves coordination—restricting and directing the flow of communications—is what impedes problem-solving. In peer groups, moreover, the free flow of communication that contributes to problem-solving also creates an informal differentiation of status as some members earn the respect and deference of others, and this differentiation, once established, creates obstacles to communication. This dilemma appears to be inherent in the conflicting requirements of coordination and problem-solving. To be sure, some types of centralized direction are more compatible with work on complex problems than others, but the fundamental dilemma posed by the need for unrestricted and for restricted communication cannot be resolved—it must be endured.

Bureaucratic Discipline and Professional Expertness. Weber's approach to the study of administration fails to distinguish the principles that govern bureaucratic organizations from professional principles, as both Parsons and Gouldner have emphasized.[36] To be sure, these two sets of principles have much in common. Both require that decisions be governed by universalistic standards independent of any personal considerations in the particular cases handled. The orientations of both professionals and bureaucrats are expected to be impersonal and detached, a principle designed to facilitate rational judgment. Both bureaucracy and professionalism are marked by specialized competence based on technical training and limit the officials's or professional's authority to a specialized area of jurisdiction. Both professionals and bureaucrats occupy an achieved rather than ascribed status, with the selection of personnel governed by such performance criteria as competence and training. These are important similarities, but they should not be allowed to obscure the equally important differences between the two.

The first difference between the organizing principles of a profession and those of a bureaucracy is that the professional is bound by a norm of service and a code of ethics to represent the welfare and interests of his clients, whereas the bureaucrat's foremost responsibility is to represent and promote the interests of his organization. Only in the case of service organizations do the ultimate objectives of serving clients

36 Talcott Parsons, "Introduction" to Max Weber, *The Theory of Social and Economic Organization,* A. M. Henderson and Talcott Parsons (trans.) and Talcott Parsons (ed.), Glencoe, Ill.: Free Press and Falcon's Wing Press, 1947, pp. 58–60; and Gouldner, *Patterns of Industrial Bureaucracy, op. cit.,* pp. 22–24.

and serving the organization coincide, and even here the specific immediate objectives often conflict. For a service organization is oriented to serving the collective interests of its entire clientele, which demands that the interests of some clients may have to be sacrificed to further those of the majority or of future clients, while the distinctive feature of the professional orientation is that each client's interests reign supreme and must not be sacrificed for the sake of the welfare of other clients.

A second basic difference concerns the source of authority. The bureaucratic official's authority rests on a legal contract backed by formal sanctions, but the professional's authority is rooted in his acknowledged technical expertness. Although some technical competence may be required for performing the duties of a customs official, it is not this skill but his legal status that authorizes the customs inspector to decide whether goods can be imported duty-free or not. An individual is legally obligated to submit to the authority of the policeman, whatever he thinks of his decision, but the same person submits to the authority of his doctor because, and only if, he acknowledges that the doctor has the technical knowledge to determine whether he should have surgery, medicine, or neither.

A third difference, related to the foregoing, is that the bureaucrat's decisions are expected to be governed by disciplined compliance with directives from superiors, whereas the professional's are to be governed by internalized professional standards. To be sure, superiors may be more highly qualified in a field than their subordinates. The crucial problem, however, is that bureaucratic management must base its decisions in part on administrative considerations, which often conflict with purely professional considerations.

Finally, the differences between the two systems are reflected in the locus of the last court of appeal in case of disagreement. When a decision of a bureaucrat is questioned, the final judgment of whether he is right or not is a prerogative of management, but when a decision of a professional is questioned, the right of reviewing its correctness is reserved to his professional colleague group. The actions of the professional expert, therefore, are under the ultimate control of his peers who have the same specialized skills as he, whereas control over the bureaucrat's action is exercised by superiors in the organization whose technical skills tend to differ from his. One complains to the medical society or to the bar association about a physician's or a lawyer's actions, and there his professional colleagues will judge whether or not the complaint is justified; but one complains to a mechanic's boss about a mechanic's actions, and the boss who judges the mechanic is typically not an expert mechanic himself.

With increasing numbers of professionals being employed in

bureaucratic settings, much attention has been directed toward examining conflicts between the demands of the administrative organization and those of professional standards. These conflicts usually find expression in contrasting orientations of employees; some adopt management as their major reference group, and others, their professional colleagues. The significance of this difference is indicated by the fact that studies of professionals or semiprofessionals in formal organizations have consistently found that the conflict between bureaucratic and professional orientation is a fundamental issue. Hughes reports conflicts between itinerants and the homeguard in numerous work settings;[37] Francis and Stone emphasize the distinction between a service and a procedure orientation in their study of a public employment agency;[38] and Gouldner focuses on the contrast between cosmopolitan and local orientations in his study of a college faculty.[39] Our research, too, found that semiprofessional workers in a public assistance agency could be differentiated on the basis of whether their orientation was confined to the organization or extended to the profession of social work. Those oriented to their profession tended to be less attached to the welfare agency, more critical of its operations—particularly of service to clients—and less confined by administrative procedure. Although a professional orientation motivates a person to do better work in terms of professional standards, it also gives him a basis for ignoring administrative considerations and thus may lead to poorer performance in terms of the standards of the organization. Thus, professionally oriented caseworkers were more apt than others to fail to visit their clients on schedule.

Research on production organizations in widely different social contexts indicates that a rational organization for the collective pursuit of formally established goals may exist whether or not the specific mechanism employed for this purpose is a bureaucratic structure. Stinchcombe presents a comparative analysis of construction and mass-production industries in our highly complex and industrialized society.[40] Udy reports a quantitative investigation of rudimentary production organizations in a large number of simple, non-Western societies.[41] Despite the great difference in source materials, the two studies arrive at essentially the same conclusion. The findings of both indicate that a

[37] Everett C. Hughes, *Men and Their Work,* Glencoe, Ill.: Free Press, 1958, pp. 31, 129–130, 136.

[38] Roy G. Francis and Robert C. Stone, *Service and Procedure in Bureaucracy,* Minneapolis: University of Minnesota Press, 1956.

[39] Alvin W. Gouldner, "Cosmopolitans and Locals," *Administrative Science Quarterly,* 2 (1957), pp. 281–306.

[40] Arthur L. Stinchcombe, "Bureaucratic and Craft Administration of Production," *Administrative Science Quarterly,* 4 (1959), pp. 168–187.

[41] Stanley H. Udy, Jr., " 'Bureaucracy' and 'Rationality' in Weber's Theory," *American Sociological Review,* 24 (1959), pp. 791–795.

rational formal organization may be but is not necessarily bureaucratic. Specifically, the fact that an organization is governed by such rational principles as specialization, rewards for performance, and contractual agreements is independent of the existence of a bureaucratic structure, that is, a hierarchy of authority and an administrative apparatus. Stinchcombe concludes that the professionalized labor force in the construction industry serves as an alternative to bureaucratization for assuring rational production, because seasonal fluctuations in this industry make it impractical to maintain continuous bureaucratic organizations. Seasonal variation, however, is not the only condition that encourages employment of a professional labor force; another is the complexity of the services to be performed. When the over-all responsibility of the organization cannot be broken down into fairly routine specialized tasks—as exemplified by organizations responsible for research, the care of the ill, and casework service—expert judgments of professionals rather than disciplined compliance with the commands of superiors must govern operations in the interest of efficiency.

Professional expertness and bureaucratic discipline may be viewed as alternative methods of coping with areas of uncertainy. Discipline does so by reducing the scope of uncertainty; expertness, by providing the knowledge and social support that enable individuals to cope with uncertainty and thus to assume more responsibility. The dilemma, however, remains and, indeed, affects wider and wider circles as the number of people subject to both these conflicting control mechanisms grows, since the work of professionals is increasingly carried out in bureaucratic organizations, and since operations in bureaucracies seem to become increasingly professionalized, modern warfare being a conspicuous example.

Managerial Planning and Initiative. The need for centralized planning and individual initiative poses a third dilemma for formal organizations—or, perhaps more correctly, a third manifestation of the basic dilemma between order and freedom.[42] Notwithstanding the importance of free communication, freedom to follow one's best professional judgment, and conditions permitting the exercise of initiative, effective coordination in a large organization requires some centralized direction. But the assumption that managerial coordination necessitates control through a hierarchy of authority is questionable, since it can be and often is achieved by other methods, notably through various types of impersonal mechanisms of control designed by management.

The assembly line is such an impersonal mechanism through which managerial planning effects coordination of the production proc-

42 See the discussion of the dilemma between bureaucracy and enterprise, as he calls it, by Marshall Dimock, *Administrative Vitality,* New York: Harper, 1959.

esses without the use of directives that are passed down the hierarchy. As a matter of fact, the impersonal constraints exerted on operators tend to reverse the flow of demand in the hierarchy. Since the moving line makes most of the demands on workers, the role of the foreman is changed from one who primarily makes demands on workers to one who responds to their demands for help and assistance, and similar changes occur on higher levels. There is centralized direction, but it is not attained through commands transmitted down the hierarchy.

Performance records are another impersonal mechanism of control, one suitable for controlling nonmanual as well as manual tasks. The regular evaluation of employee performance on the basis of quantitative records of accomplished results exerts constraints that obviate the need for routine supervisory checking. Performance records, like the assembly line, reverse the flow of demand in the organization and cast the supervisor in the role of adviser and helper to workers rather than in the role of a person who makes continual demands on them. This evaluation system also facilitates coordination, since it centralizes the direction of operations in the hands of the higher managers who design the records.

Both performance records and assembly lines minimize reliance on hierarchical authority and discipline to control operations and, therefore, improve relations between supervisors and subordinates. However, there is an important difference between these two mechanisms. Assembly-line production reduces the discretion workers can exercise and hence lowers their work satisfaction. In contrast, evaluation of performance on the basis of a quantitative record of results achieved increases the discretion employees are allowed to exercise and thus raises their work satisfaction.

We had expected that automation would be an impersonal control mechanism more similar in its consequences to performance records than to the assembly line. We further anticipated that most workers in automated plants, where routine tasks are performed by machines, would be technical experts engaged in maintenance and trouble-shooting, and that they, consequently, would enjoy more discretion and have higher work satisfaction. The surprising findings of studies conducted in automated organizations by Walker, Faunce, and Mann and Williams is that the average level of skill and responsibility of workers was not superior to the level that had existed prior to automation.[43] The discretion permitted workers had not been increased. Indeed, in the automated factory studied by Faunce, supervision was closer than on

[43] Walker, *op. cit.*; William A. Faunce, "Automation in the Automobile Industry," *American Sociological Review*, 23 (1958), pp. 401–407; and Floyd C. Mann and Lawrence K. Williams, "Observations of the Dynamics of a Change to Electronic Data-Processing Equipment," *Administrative Science Quarterly*, 5 (1960), pp. 217–256.

the assembly line, because foremen were concerned with preventing costly machine breakdowns. Since automation removed some of the higher positions in the organization as well as some lower ones, it reduced chances for advancement, a situation which was a source of considerable dissatisfaction. It appears that automated plants have not yet reorganized their work processes to take full advantage of the technological innovations. This reorganization would require, in our opinion, the training or recruitment of expert mechanics and the redesigning of the division of labor to include minor machine maintenance in the duties of operators. Under these conditions, machine breakdowns, or the impending danger of them, would not lead to closer supervision as it did in the plants studied, and the highly technical operations would permit the exercise of considerable discretion.

It is conceivable that union pressure to increase wage rates on automated jobs will force management to institute such a reorganization. Higher labor costs constrain management to attempt to improve productivity, and one means for accomplishing this improvement is through further automation that eliminates routine jobs. The remaining highly paid workers could be held responsible for acquiring the skills needed for the maintenance functions now discharged by foremen or specialists. Such changes would give them more discretion, lessen the need for close supervision, and thus probably raise work satisfaction. These predictions are in line with Melman's conclusion that union pressures and high wages have induced management to introduce technological innovations more rapidly than would otherwise have been the case.[44] Such a professionalization of the labor force might also require a reorganization of the reward system, since piece rates do not furnish incentives suited for professionalized tasks. Even in the semi-automated department studied by Walker, where tasks were far from professionalized, and where workers were quite satisfied with their rate of pay, there was much dissatisfaction with the piece-rate system for failing to take mental work and judgment into account. A reward system that emphasizes advancement chances rather than immediate earnings and evaluation of results rather than sheer productivity would seem to furnish more effective incentives for professionalized tasks.

Managerial planning of the production process and a professionalized labor force that can exercise initiative and is motivated to do so by opportunities for advancement would sharply reduce the need for hierarchical supervision and control through directives passed down the pyramid of authority. Indeed, coordination appears to be achieved frequently through centralized planning and by means of direct communication between responsible managers (as in Simpson's spinning depart-

[44] See Seymour Melman, *Decision-making and Productivity*, Oxford, England: Basil Blackwell, 1958, pp. 105–106, 141–143.

ment cited in Chapter VII) rather than through the cumbersome process of passing messages up and down the hierarchy. But our suggestion that managerial planning interferes less with the exercise of initiative than hierarchical authority is not meant to imply that the dilemma between managerial control and initiative is resolved. The best that can be hoped for, as Bendix has suggested, is that

> . . . the employees of all ranks in industry and government strike a balance between compliance and initiative, that they temper their adherence to formal rules by a judicious exercise of independent judgement and that they fit their initiative into the framework of the formal regulation.[45]

But even this best is too much to expect. For this balance is continually disrupted by the need for more order on the one hand and the need for more freedom on the other.

Dialectical Processes of Change

The conception of dilemma directs attention to the inevitability of conflict and change in organizations. Mary Parker Follett, an astute observer of administrative practice, has noted: "When we think that we have *solved* a problem, well, by the very process of solving, new elements or forces come into the situation and you have a new problem on your hands to be solved."[46] The innovations instituted to solve one problem often create others because effectiveness in an organization depends on many different factors, some of which are incompatible with others; hence, the dilemma. The very improvements in some conditions that further the achievement of the organization's objectives often interfere with other conditions equally important for this purpose. A by now familiar example is that hierarchical differentiation promotes coordination but simultaneously restricts the communication processes that benefit decision-making.

New problems are internally generated in organizations in the process of solving old ones. However, the experience gained in solving earlier problems is not lost but contributes to the search for solutions to later problems. These facts suggest that the process of organzational development is dialectical—problems appear, and while the process of solving them tends to give rise to new problems, learning has occurred

45 Reinhard Bendix, "Bureaucracy," *American Sociological Review,* 12 (1947), p. 503.

46 Mary Parker Follett, "The Process of Control," Luther Gulick and L. Urwick (eds.), *Papers on the Science of Administration,* New York: Institute of Public Administration, 1937, p. 166 (italics in original).

which influences how the new challenges are met.[47] Consequently, effectiveness in an organization improves as a result of accumulated experience. These dialectical processes are illustrated by the introduction of assembly-line production. This new production method raised productivity and effected centralized control and coordination without the need for hierarchical directives. However, by routinizing tasks and lowering work satisfaction, the assembly line created problems of absenteeism and turnover—problems that were particularly serious given the interdependence of operations on the assembly line. Management had succeeded in solving one set of problems, but the mechanism by which they were solved produced new problems which were quite different from those that had existed in earlier stages of mechanization. Contrary to our expectations, the introduction of automation has not yet met the problems created by monotonous tasks and low work satisfaction. But should these problems be solved through a reorganization of the work force that requires operators to assume more responsibility, as we have suggested, management would no doubt again be faced with new difficulties. For example, increased responsibility and discretion in performing complex, interdependent tasks might engender anxieties over decision-making which would impede effective performance, and these new problems would require management to devote attention to developing mechanisms that reduce such anxieties.

Conflicts of interest between various groups or persons in the organization are another source of dialectical change. What constitutes satisfactory adjustment for one group may be the opposite for another, since different interests serve as the criteria of adjustment. Thus, when the efforts of managers are judged by the results they achieve and they are given freedom to exercise responsibility and initiative in achieving them, conflicts between them are likely to ensue. For each manager will seek to promote the interests and expand the jurisdiction of his department, and his endeavors will bring him into conflict with others who have staked out the same claims. Compromises will be reached and coalitions will be formed, but since the responsibilities and interests of the managers continue to differ, new conflicts are apt to arise as changing conditions produce new challenges. Moreover, as various occupational subgroups in the organization try to improve their economic position, their interests may come into conflict, particularly if the success of one group upsets the existing status hierarchy and motivates the others it has displaced to recoup their advantage. Conflicts of interest are most conspicuous in the relation between union and management. The union is interested in obtaining higher wages and better working

[47] If we classify problems into dichotomies or other very broad categories, it inevitably seems as if the same ones recur, simply because all new ones are put into one of the few existing categories.

conditions, while management is interested in lowering costs and improving productivity. Collective bargaining furnishes mechanisms for resolving issues, but the conflicting interests generate new ones. Thus, management introduces new machines in an attempt to improve efficiency, disturbing the existing adjustment and producing a variety of difficulties with which the union has to deal. Similarly, once workers have attained the right to collective bargaining, they use it to fight for pensions and other fringe benefits, thereby creating new problems for management.

Another source of disruption and change is turnover in personnel. Valuable experience is lost as older workers are replaced by new trainees, and social ties are disrupted by transfers and loss of personnel. As we have seen, the methods available to a new manager in discharging his responsibilities are dependent in part on those of his predecessor. If the latter commanded the loyalty of subordinates, the successor will find it difficult to do so and be constrained to resort to bureaucratic methods, whereas the successor to an authoritarian bureaucrat will find it advantageous to use more informal managerial practices. Again we see that organizational developments alternate in direction in a dialectical pattern. The succession of goals leads to such an alternating pattern of change in the relations between organizations. Once earlier objectives are achieved, management seeks new objectives and by doing so disturbs the existing equilibrium in the network of organizations. The dominance of one organization in a sector restores order as former competitors become exchange partners, but further power struggles are stimulated by a further succession of goals as groups of sellers start to compete with groups of buyers for dominant power over a set of related markets.

In mutual-benefit associations, there is still another source of dialectical change. These organizations are subject to conflicts that arise from the dilemma posed by their twofold formal purpose. One purpose, just as in the case of other organizations, is the effective accomplishment of the specific objectives of the organization—for example, improving employment conditions in the case of unions. But another distinctive purpose of these associations is to provide their members with a mechanism for arriving at agreements on their common objectives. For to serve the interests of its members a mutual-benefit association must furnish mechanisms for ascertaining what their collective objectives are as well as mechanisms for implementing them, and the ascertaining of objectives requires democratic self-government and freedom of dissent. Endeavors to attain one of these purposes frequently impede the attainment of the other. In the interests of effective accomplishment of union objectives, as Michels has pointed out, democratic processes are often set aside. Conversely, preoccupation with democratic

self-government and freedom of dissent may interfere with efforts to implement the common objectives. But the study by Lipset and his colleagues shows that a strong union which has accomplished some of its specific objectives can and sometimes does turn its attention and energy to maintaining internal democracy.

Democratic societies are in this respect organized like mutual-benefit associations. They have the double purpose of remaining strong enough to survive and yet maintaining the freedoms that permit the democratic establishment of common objectives. Under current conditions in the world, the issue of promoting national security and strength versus preserving civil liberties and freedom of dissent poses the dilemma most sharply. No final solution is possible for this dilemma. Indeed, attempts finally to resolve it tend to sacrifice one purpose for the other and thus endanger the very nature of democratic societies. For we surely need to survive in order to preserve our democratic institutions, but we just as surely do not want to survive at the cost of losing our freedom.

APPENDIX

Description and Comparison of the Two Welfare Agencies

In 1957 Blau conducted a study of a city welfare agency in a large metropolis,[1] which we call "City Agency." In 1959 Scott carried out a study of a county welfare agency in a smaller urban center in a neighboring state,[2] which we refer to as "County Agency." In part the two studies were concerned with different problems; for example, the orientation of caseworkers was a main focus of Scott's investigation, while Blau's centered more on supervision. But in part the second study replicated the first, notably the material on relations among colleagues, providing opportunities for systematic comparison. We want to say a few words about data collection and then describe the two agencies to point out the similarities that make them comparable as well as the differences that furnish the basis for the comparative analysis.

Both investigations dealt primarily with the professional staffs of the agencies rather than the clerical or purely administrative employees. Data were collected in City Agency by systematic as well as unstructured observation, interviewing, and analysis of records and documents. Of a total of 70 caseworkers, 60 (86 per cent) in two field sections consented to be interviewed, as did 10 of the 12 supervisors directly in charge of them (all 12 supervisors as well as many workers and most of the higher supervisory staff had been informally interviewed). The systematic interviews were conducted after the period of observation either in a private room at the agency or at the worker's home, as he preferred. The study of County Agency employed unstructured and systematic observation, interviewing, a self-administered questionnaire, and the analysis of records and documents. Systematic data on workers were collected primarily by questionnaire subsequent to the period of observation. Of the 92 caseworkers, 90 (98 per cent) returned a lengthy questionnaire. Interviews were conducted with 11 of the 12 first-line supervisors during office hours.

Both organizations were public agencies charged with the respon-

[1] Philip M. Marcus served as research assistant for this study. Funds for it were made available by the Social Science Research Committee of the University of Chicago.

[2] A predoctoral training fellowship from the Social Science Research Council provided financial support for this study.